

Danish Shakil

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Mechanical Design · CAD · Simulation

PROFESSIONAL SUMMARY

Mechanical engineering student turning technical specifications into manufacturable CAD — from a client enclosure design for a US trailer maker to a titanium aerospace bracket optimized under FEA. Skilled in 3D modeling, mechanical simulation, and interpreting technical specifications to deliver client-facing design work, including a completed manufacturing-process project for British American Tobacco.

EDUCATION

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI) Expected 2027

Bachelor of Science in Mechanical Engineering

CGPA: 3.29/4.00

TECHNICAL SKILLS

- **CAD & Modeling:** SolidWorks (CSWA Certified), Blender
- **Engineering & Analysis:** ANSYS, COMSOL Multiphysics, MATLAB, Simulink, Dynacam, Linkages
- **Robotics & Embedded:** ROS 2, RViz2, Proteus, Arduino IDE
- **Manufacturing:** 3D printing, rapid prototyping, additive manufacturing
- **Core Competency:** Machine design to Shigley's standards, GD&T, topology optimization, geometric inverse kinematics

PROFESSIONAL EXPERIENCE

British American Tobacco (BAT)

Jun 2026 – Jul 2026

Mechanical Design Intern — Automated Tobacco Reclaimer System

- Designed and mapped an automated tobacco reclaimer system as part of a manufacturing internship, aimed at optimizing primary production operations.
- Produced technical design reports and detailed cross-departmental process flow diagrams for engineering and operations stakeholders.

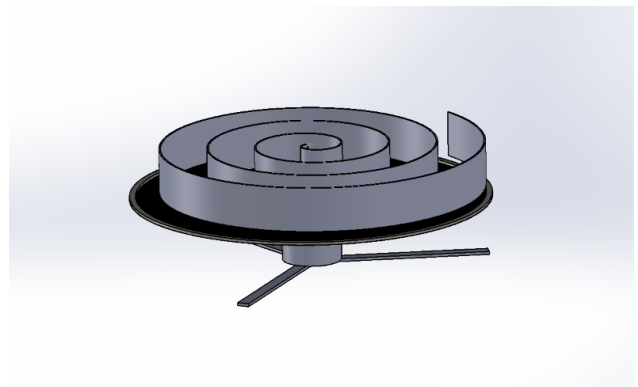
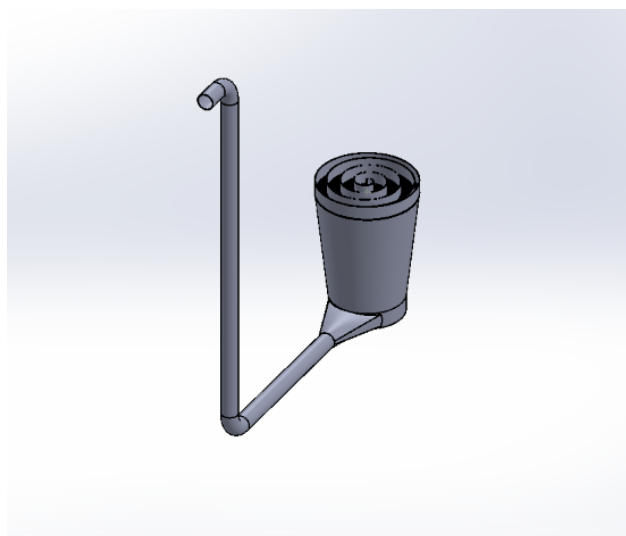


Figure 1: Left: Process flow diagram of the automated tobacco reclaimer system. Right: Technical design report visualization.

Bigfoot Trailers

Aug 2025

CAD Design Contractor — Yeti 24 Product Enclosure Design

- Delivered the complete 3D CAD model for a Yeti-branded product enclosure for a US-based trailer manufacturer, working remotely to client specifications.
- Ensured designs strictly adhered to client requirements, optimizing structure for manufacturability and aesthetic appeal.

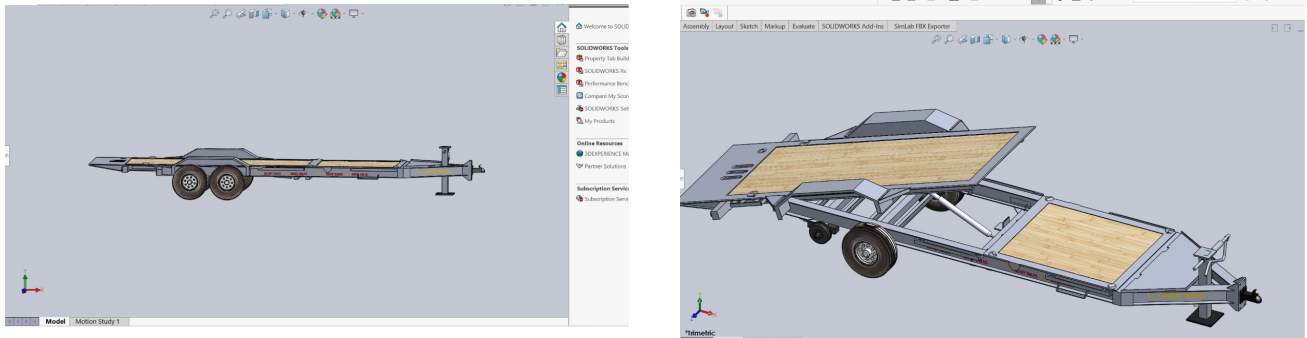


Figure 2: Left: Yeti 24 product enclosure full exterior CAD render. Right: Detailed section view showcasing internal mounting points.

ACADEMIC & COMPETITION PROJECTS

RoboCup Small Size League Robot

Dec 2025 – May 2026

Autonomous Soccer Robot, Maxon-Motor Drivetrain

- Developed 3D CAD assemblies for an autonomous soccer robot equipped with Maxon motors.
- Designed and modeled a specialized mechanical dribbler and kicker mechanism in SolidWorks, optimizing packaging constraints and internal clearances.
- Managed iterative design revisions for rapid prototyping and executed 3D printing of complex structural components.

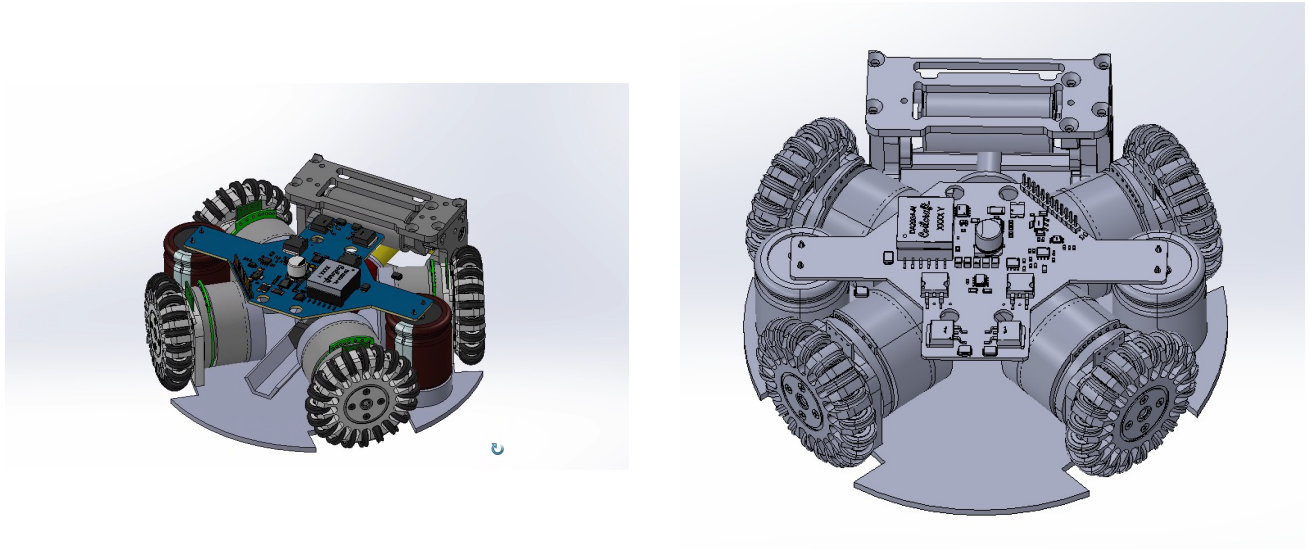


Figure 3: Left: Overall CAD assembly of the autonomous soccer robot. Right: Detailed view of the mechanical dribbler and kicker mechanism.

Heavyweight Combat Robot

Jul 2024 – May 2025

60 kg Vertical Spinner

- Executed structural calculations, material selection, and force analysis to ensure design integrity under heavy impact.
- Developed the full 3D CAD assembly, ensuring all internal mechanisms met dimensional and weight constraints.
- Transitioned CAD models into physical builds through hands-on fabrication, machining, and rapid prototyping.



Figure 4: Left: Structural layout and weapon integration of the 60 kg Heavyweight Combat Robot[cite: 5]. Right: Top-down CAD view of the robot chassis and internal component placement[cite: 5].

Ti-6Al-4V Titanium Alloy Structural Bracket

- Executed a topology optimization study using ANSYS to lightweight a standard Ti-6Al-4V aerospace bracket while preserving its load-bearing integrity[cite: 5].
- Defined the design space, non-design regions (mounting holes), and applied multi-axis loading conditions simulating in-flight forces (e.g., 10 kN static load, 5G vibrational load)[cite: 5].
- Iteratively refined the generated organic geometry to enforce additive manufacturing constraints (e.g., overhang angles, minimum feature sizes)[cite: 5].
- Achieved a 32% reduction in overall component mass while maintaining a structural Factor of Safety (FoS) > 1.5, verifying the final design via a secondary Von Mises stress analysis[cite: 5].

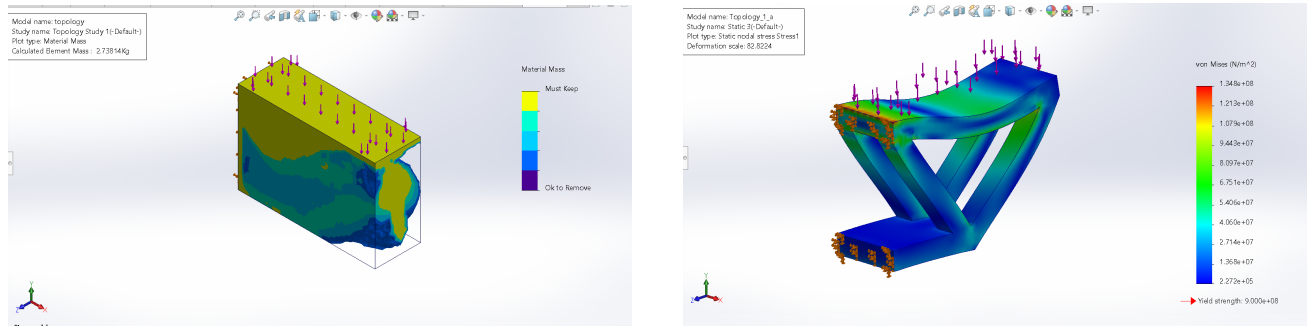


Figure 5: Left: ANSYS Von Mises stress distribution of the baseline bracket geometry[cite: 5]. Right: Stress contour plot of the topology-optimized, lightweighted design[cite: 5].

PROFESSIONAL CERTIFICATIONS

- **Certified SOLIDWORKS Associate (CSWA)** — Dassault Systèmes[cite: 5]
- **SOLIDWORKS Electrical Design Certification** — Dassault Systèmes[cite: 5]
- **Additive Manufacturing Certification**[cite: 5]